

Arnab Roy

Overview

I design and productize cryptographic protocols for decentralized systems, with applications to finance and digital assets. My research spans zero-knowledge proofs, blockchain, secure communication, biometric authentication, data privacy, post-quantum cryptography, formal verification, hardware enclaves, machine learning, and quantum-inspired algorithms. I have been a core contributor to several products, 50+ research publications, and 40+ patents.

Education

Ph.D. in Computer Science, 2009
Stanford University

B.Tech. and M.Tech. in Computer Science and Engineering, 2004
Indian Institute of Technology Kharagpur

Employment

2022 - :	Research Scientist Mysten Labs, Palo Alto, CA
2021 - 2022 :	Research Scientist Manager Meta Platforms Inc, Menlo Park, CA
2012 - 2021:	Senior Research Manager (2021) and Global head of cryptography research and blockchain OSS (2020-21) Fujitsu Research of America, Sunnyvale, CA
2010 - 2012:	Post Doctoral Researcher IBM Thomas J. Watson Research Center, Hawthorne, NY

Awards and Honours

Young Alumni Achiever Award 2019, IIT Kharagpur.

Siebel Scholar Class of 2006, Stanford Computer Science.

Prime Minister of India Gold Medal 2004, IIT Kharagpur.

Top scoring paper, PKC 2018, a premier cryptography conference.

Best Paper Award, AsiaCrypt 2013, a flagship conference in cryptography.

Co-Chair, NIST Big Data Security and Privacy Subgroup, 2013-20.

Product Contributions

At Mysten Labs, I have been a core contributor to several products at the intersection of cryptography, distributed systems, language design, and formal verification, including:

1. zkLogin: Wallets based on email and social logins. Co-inventor, prototype and product collaborator, lead for the MPC ceremony, and co-lead for publications at CCS, SBC, and RWC.
2. Seal: Decentralized secrets management governed by Move policies. Design coauthor, whitepaper lead with cryptographic formalization and auditing.
3. Hashi: Decentralized Bitcoin collateralization primitive. Whitepaper lead with cryptographic formalization, and design, and prototyping for novel research components.
4. Confidential Transfers: Encrypted balances with auditability. Co-designer and whitepaper lead with cryptographic formalization, as well as lead for the design and prototyping of upcoming novel research optimizations accompanied with Lean proofs.

Publications

Conferences

1. Dan Boneh, Rohit Nema, Arnab Roy, and Ertem Nusret Tas, "Efficient Batch Threshold Encryption Using Partial Fraction Techniques". In *Proceedings of CRYPTO 2026, Science of Blockchain Conference (SBC) 2026*.
2. Charanjit S. Jutla, Rohit Nema, and Arnab Roy, "Partial Fraction Techniques for Cryptography". In *Proceedings of EuroCrypt 2026, Science of Blockchain Conference (SBC) 2026*.
3. Foteini Baldimtsi, Kostas Kryptos Chalkias, Arnab Roy, and Mahdi Sedaghat. "Post-Quantum Readiness in EdDSA Chains". In *Proceedings of FC 2026*.
4. Kostas Kryptos Chalkias, Charanjit S. Jutla, Jonas Lindstrøm, Varun Madathil, and Arnab Roy. "A Linear Operator Framework for Improved Polynomial Division in Cryptography". In *Proceedings of AsiaCCS 2026*.
5. Kostas Kryptos Chalkias, Deepak Maram, Arnab Roy, Joy Wang, and Aayush Yadav. "Zero-knowledge Authenticator for Blockchain: Policy-private and Obliviously Updateable". In *Proceedings of AFT 2025*.
6. Foteini Baldimtsi, Kostas Kryptos Chalkias, Yan Ji, Jonas Lindstrøm, Deepak Maram, Ben Riva, Arnab Roy, Mahdi Sedaghat, and Joy Wang. "zkLogin: Privacy-Preserving Blockchain Authentication with Existing Credentials", In *Proceedings of CCS 2024, Science of Blockchain Conference (SBC) 2024*, and *Real World Cryptography (RWC) 2025*.
7. Miranda Christ, Foteini Baldimtsi, Kostas Kryptos Chalkias, Deepak Maram, Arnab Roy, and Joy Wang. "SoK: Zero-Knowledge Range Proofs", In *Proceedings of AFT 2024*.
8. Kostas Kryptos Chalkias, Jonas Lindstrøm, Deepak Maram, Ben Riva, Arnab Roy, Alberto Sonnino, and Joy Wang. "Fastcrypto: Pioneering Cryptography Via Continuous Benchmarking", In *The Twelfth International Workshop on Load Testing and Benchmarking of Software Systems (LTB 2024)*.
9. Foteini Baldimtsi, Konstantinos Chalkias, Francois Garillot, Jonas Lindstrøm, Ben Riva, Arnab Roy, Mahdi Sedaghat, Alberto Sonnino, Pun Waiwitikhith, and Joy Wang. "Subset-optimized BLS Multi-signature with Key Aggregation", In *Proceedings of FC 2024*.

10. Rex Fernando, and Arnab Roy. "Poster: WIP: Account ZK-Rollups from Sumcheck Arguments", In *Proceedings of CCS 2023*.
11. Donald Beaver, Konstantinos Chalkias, Mahimna Kelkar, Lefteris Kokoris-Kogias, Kevin Lewi, Ladi de Naurois, Valeria Nikolaenko, Arnab Roy, and Alberto Sonnino. "STROBE: Streaming Threshold Random Beacons", In *Proceedings of AFT 2023*.
12. Samuel Jaques, Hart Montgomery, Razvan Rosie, Arnab Roy. "Time-Release Cryptography from Minimal Circuit Assumptions." In *Proceedings of INDOCRYPT 2021*.
13. Eldan Cohen, Avradip Mandal, Hayato Ushijima-Mwesigwa, Arnab Roy. "Unified Clustering and Outlier Detection on Specialized Hardware." In *Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing (IEEE ICASSP 2021)*.
14. Eldan Cohen, Avradip Mandal, Hayato Ushijima-Mwesigwa, Arnab Roy. "Ising-Based Consensus Clustering on Specialized Hardware." In *Proceedings of Advances in Intelligent Data Analysis (IDA 2020)*.
15. Avradip Mandal, Arnab Roy, Sarvagya Upadhyay, and Hayato Ushijima-Mwesigwa. "Compressed quadratization of higher order binary optimization problems." In *Proceedings of the 17th ACM International Conference on Computing Frontiers (ACM CF 2020)*.
16. Masayuki Abe, Charanjit S. Jutla, Miyako Ohkubo, Jiaxin Pan, Arnab Roy and Yuyu Wang, "Shorter QA-NIZK and SPS with Tight Security", In *Proceedings of AsiaCrypt 2019*.
17. Navid Alapati, Hart Montgomery, Sikhar Patranabis and Arnab Roy, "Minicrypt Primitives with Algebraic Structure and Applications", In *Proceedings of EuroCrypt 2019*.
18. Masayuki Abe, Charanjit S. Jutla, Miyako Ohkubo and Arnab Roy, "Improved (Almost) Tightly-Secure Simulation-Sound QA-NIZK with Applications", In *Proceedings of AsiaCrypt 2018*.
19. Charanjit S. Jutla and Arnab Roy, "Smooth NIZK Arguments", in *Proceedings of TCC 2018*.
20. Avradip Mandal, John C. Mitchell, Hart Montgomery and Arnab Roy, "Data Oblivious Genome Variants Search on Intel SGX", In *Proceedings of Data Privacy Management 2018*.
21. Sam Kim, Kevin Lewi, Avradip Mandal, Hart Montgomery, Arnab Roy and David Wu, "Function-Hiding Inner Product Encryption is Practical", In *Proceedings of SCN 2018 and presented at ArtCrypt 2016*.
22. Charanjit S. Jutla, Miyako Ohkubo and Arnab Roy, "Improved (Almost) Tightly-Secure Structure-Preserving Signatures", in *Proceedings of PKC 2018*. **[Top scoring paper]**
23. Avradip Mandal, John C. Mitchell, Hart Montgomery and Arnab Roy, "Privacy for Targeted Advertising", in *Proceedings of IEEE CNS-SPA 2017*.
24. Charanjit S. Jutla and Arnab Roy, "Improved structure preserving signatures under standard bilinear assumptions", in *Proceedings of PKC 2017*.
25. Avradip Mandal, Arnab Roy and Masaya Yasuda, "Comprehensive and Improved Secure Biometric System using Homomorphic Encryption", In *Proceedings of DPM 2015*.
26. Avradip Mandal and Arnab Roy, "Relational Hash: Probabilistic Hash for Verifying Relations, Secure against forgery and More", in *Proceedings of CRYPTO 2015*.
27. Charanjit S. Jutla and Arnab Roy, "Dual-System Simulation-Soundness with Applications to UC-PAKE and More", In *Proceedings of AsiaCrypt 2015*.

28. J. Hartloff, Avradip Mandal and Arnab Roy, "Privacy Preserving Technique for Set-Based Biometric Authentication using Reed-Solomon Decoding", In *Proceedings of BIOSIG 2015*.
29. Charanjit S. Jutla and Arnab Roy, "Switching Lemma for Bilinear Tests and Constant-size NIZK proofs for Linear Subspaces", in *Proceedings of CRYPTO 2014*.
30. Charanjit S. Jutla and Arnab Roy, "Shorter Quasi-Adaptive Proofs for Linear Subspaces", in *Proceedings of AsiaCrypt 2013*. [Best Paper Award]
31. Charanjit S. Jutla and Arnab Roy, "Decision Procedures for Simulatability", in *Proceedings of ESORICS 2012*: 573-590.
32. Charanjit S. Jutla and Arnab Roy, "Relatively-Sound NIZKs and Password-Based Key-Exchange". in *Proceedings of Public Key Cryptography (PKC) 2012*: 485-503.
33. Ran Canetti, Suresh Chari, Shai Halevi, Birgit Pfitzmann, Arnab Roy, Michael Steiner and Wietse Venema, "Composable Security Analysis of OS Services", *Proceedings of Applied Cryptography and Network Security (ACNS) 2011*.
34. Hang Zhao, Jorge Lobo, Arnab Roy and Steven Bellovin, "Policy Refinement of Network Services for MANETs", *Proceedings of 12th IFIP/IEEE International Symposium on Integrated Network Management (IM) 2011*.
35. Yuri Gurevich and Arnab Roy, "Operational Semantics for DKAL: Application and Analysis", in *Proceedings of TrustBus 2009*: 149-158.
36. John C. Mitchell, Arnab Roy, Paul Rowe and Andre Scedrov, "Analysis of EAP-GPSK Authentication Protocol", in *Proceedings of Applied Cryptography and Network Security (ACNS), 6th International Conference*, pp. 309-327, June 2008.
37. Arnab Roy, Anupam Datta and John C. Mitchell, "Formal Proofs of Cryptographic Security of Diffie-Hellman-based Protocols", in *Proceedings of 3rd Symposium on Trustworthy Global Computing (TGC)*, November 2007.
38. Arnab Roy, Anupam Datta, Ante Derek and John C. Mitchell, "Inductive Proofs of Computational Secrecy", in *Proceedings of 12th European Symposium On Research In Computer Security (ESORICS)*, September 2007.
39. Subrat K. Panda, Arnab Roy, Partha P. Chakrabarti, Rajeev Kumar, "Simulation Based Verification using Temporally Attributed Boolean Logic.", in *Proceedings of VLSI Design 2007*: 57-62
40. Arnab Roy, Anupam Datta, Ante Derek and John C. Mitchell, "Inductive Trace Properties for Computational Security", in *ACM SIGPLAN and IFIP WG 1.7 7th Workshop on Issues in the Theory of Security (WITS)*, March 2007. [Invited to special issue of *Journal of Computer Security*]
41. Arnab Roy, Anupam Datta, Ante Derek, John C. Mitchell and Jean-Pierre Seifert, "Secrecy Analysis in Protocol Composition Logic", in *Proceedings of 11th Annual Asian Computing Science Conference (ASIAN)*, December 2006.

Journal Articles

1. Navid Alamedi, Hart Montgomery, Sikhar Patranabis and Arnab Roy, "Minicrypt Primitives with Algebraic Structure and Applications", *Journal of Cryptology*, 2022.
2. Xiaoyuan Liu, Hayato Ushijima-Mwesigwa, Avradip Mandal, Sarvagya Upadhyay, Ilya Safro, Arnab Roy. "Leveraging special-purpose hardware for local search heuristics", *Comput. Optim. Appl.* 82(1): 1-29 (2022).

3. Osman Asif Malik, Hayato Ushijima-Mwesigwa, Arnab Roy, Avradip Mandal, and Indradeep Ghosh. "Binary matrix factorization on special purpose hardware." *PloS one* 16, no. 12 (2021): e0261250.
4. Charanjit S. Jutla and Arnab Roy, "Shorter Quasi-Adaptive NIZK Proofs for Linear Subspaces", *Journal of Cryptology*, 2016.
5. Arnab Roy, Anupam Datta, Ante Derek and John C. Mitchell, "Inductive Trace Properties for Computational Security", *Journal of Computer Security* 18(6): 1035-1073 (2010).
6. John C. Mitchell, Arnab Roy and Mukund Sundararajan, "An Automated Approach for Proving PCL Invariants", *Electr. Notes Theor. Comput. Sci.* 234: 93-113 (2009)
7. Subrat K. Panda, Arnab Roy, Partha P. Chakrabarti and R. Kumar, "Simulation-based verification using Temporally Attributed Boolean Logic, *ACM Trans. Design Autom. Electr. Syst.* 13(4): (2008)
8. Anupam Datta, Ante Derek, John C. Mitchell and Arnab Roy, "Protocol Composition Logic (PCL)", *Electronic Notes in Theoretical Computer Science*, Volume 172 , 1 April 2007, Pages 311-358. Computation, Meaning, and Logic: Articles dedicated to Gordon Plotkin.
9. Arnab Roy, Subrat K. Panda, Rajeev Kumar, Partha P. Chakrabarti, "A framework for systematic validation and debugging of pipeline simulators", *ACM Trans. Design Autom. Electr. Syst.* 10(3): 462-491 (2005)

Standards Publications

1. **Co-chair.** "NIST Big Data Interoperability Framework: Volume 4, Big Data Security and Privacy." (2013-18).
2. Contributor. "NIST Big Data Interoperability Framework: Volume 1, Big Data Definitions." (2013-18).
3. Contributor. "NIST Big Data Interoperability Framework: Volume 2, Big Data Taxonomies." (2013-18).
4. Contributor. "NIST Big Data Interoperability Framework: Volume 6, Reference Architecture." (2013-18).
5. Contributor. "NIST cloud computing security reference architecture (No. NIST Special Publication (SP) 500-299 (Retired Draft))." 2013.
6. **Lead author.** "Top ten big data security and privacy challenges." Cloud Security Alliance (2012).
7. Co-author. "Big Data Taxonomy". Cloud Security Alliance (2014).

Book Chapters and Theses

1. Anupam Datta, John C. Mitchell, Arnab Roy and Stephan H. Stiller, "Protocol Composition Logic", *Cryptology and Information Security Series*, Volume 5: Formal Models and Techniques for Analyzing Security Protocols, IOS Press, July 2010.
2. Arnab Roy, "Formal Proofs of Cryptographic Security of Network Protocols", PhD Thesis, Computer Science Department, Stanford University, December 2009.
3. Arnab Roy, Anupam Datta, Ante Derek, John C. Mitchell, and Jean-Pierre Seifert, "Secrecy Analysis in Protocol Composition Logic", in *Formal Logical Methods for System Security and Correctness*, IOS Press, 2008. Volume based on presentations at Summer School 2007, Formal Logical Methods for System Security and Correctness, Marktoberdorf, Germany.

Program Committees

AFT 2026, AsiaCCS 2026-27, SBC 2025-26, CESC.IO 2022, EuroCrypt 2016, IEEE Big Data 2014-15, 20, LOD (MOD) 2015-22

Patents

40+ US Patents granted on biometric authentication, secure communication, data privacy, blockchain, post-quantum cryptography, trusted execution environments, machine learning and quantum-inspired algorithms. 10+ internal technology transfers based on these patents.